



Aishu R 251001007-2025-IT ▾

**A2****Started on** Monday, 24 August 2026, 12:30 PM**State** Finished**Completed on** Monday, 24 August 2026, 12:31 PM**Time taken** 1 min 3 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findPair(int a[], int left, int right, int x, int *p, int *q)
4  {
5      int sum;
6
7      if (left >= right)
8          return 0;
9
10     sum = a[left] + a[right];
11
12     if (sum == x)
13     {
14         *p = a[left];
15         *q = a[right];
16         return 1;
17     }
18
19     if (sum < x)
20         return findPair(a, left + 1, right, x, p, q);
21
22     return findPair(a, left, right - 1, x, p, q);
23 }
24
25 int main()
26 {
27     int n, x, i;
28     int a[1000];
29     int p, q;
30
31     scanf("%d", &n);
32
33     for (i = 0; i < n; i++)
34         scanf("%d", &a[i]);
35
36     scanf("%d", &x);
37
38     if (findPair(a, 0, n - 1, x, &p, &q))
39     {
40         printf("%d\n", p);
41         printf("%d\n", q);
42     }
43     else
44     {
45         printf("No");
46     }
47
48     return 0;
49 }
50

```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 4        | 4   | ✓ |
|   | 2     | 10       | 10  |   |
|   | 4     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 14    |          |     |   |
| ✓ | 5     | No       | No  | ✓ |
|   | 2     |          |     |   |
|   | 4     |          |     |   |
|   | 6     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 100   |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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